higher education & training

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INDUSTRIAL ELECTRONICS N5

(8080175)

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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QUESTION 1: ALTERNATING-CURRENT THEORY

1.1 What are coupling capacitors used for? (2)

1.2 Draw a graphical representation of every reactance, resistance and impedance of a series-RLC-circuit versus frequency. (4)

1.3 TWO circuits A and B are connected in parallel to a 200 V, 50 Hz supply. The current taken by the combination is 15 A. Circuit A consists of a 15 Ω resistor and a 200 µf capacitor connected in series. Circuit B consists of a resistor and an inductor in series.

Calculate the following data for circuit B:

1.3.1 The current through A and B (7)

1.3.2 The impedance across B (2)

1.3.3 The reactance and resistance across B (1)

[16]

QUESTION 2: POWER SUPPLIES

2.1 The following values of a RC-filter are known: X_{C2} = 26,526 Ω; R_L = 1,2 kΩ; R = 450 Ω; r' = 1,3% and $V'_{R(RMS)}$ = 0,118 V at C_2 f = 50 Hz. The supply is rectified by a half-wave rectifier.

Calculate:

2.1.1 V_{DC} across the first capacitor (4)

2.1.2 $V_{R(RMS)+}$ across the first capacitor (3)

2.1.3 r across the first capacitor (2)

2.2 Briefly define the term *voltage regulation.* (2)

2.3 Draw a neat, labelled block diagram of a double power supply, which can supply both positive and negative voltages to operational amplifiers. (6)

[17]

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QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 3.1.1 Indicate, using TWO sketches, how the operating point on the load line of a common emitter amplifier can shift due to temperature changes. (4)

3.1.2 Briefly explain how the shifting of the operating point could be minimised. (3)

3.2 In a voltage divider biased amplifier the values of R_{B1} and R_{B2} are 74 kΩ and 7,4 kΩ respectively.

Calculate the values of thermal components R_E and C_E, if the minimum signal frequency is 50 Hz and β = 200. (4)

3.3 A fixed biased voltage amplifier has the following information:

h_{ie} = 1 kΩ	h_{oe} = 25 µA/V	h_{re} = 2,5 x 10^{-4}
h_{fe} = 50	R_B = 200 kΩ	R_C = 2,2 kΩ

Calculate according to the PRECISION method the following:

3.3.1 Input impedance of the transistor (3)

3.3.2 Voltage gain of the amplifier (2)

[16]

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 Name TWO common mode signals that may appear in differential amplifiers. (2)

4.2 Draw the pin outlay and functional diagram of a 741-operational amplifier in an 8-pin dual in-line package. (4)

4.3 Calculate the value of the feedback resistor in a practical differentiator for a 2 kHz signal for a chosen capacitor with a value of 2 µf. (3)

4.4 Draw a neat, labelled circuit diagram of an operational amplifier which is used for impedance matching. (2)

[11]

QUESTION 5: INTEGRATED CIRCUITS

5.1 Show, by means of a neat, labelled sketch, how the terminals of a 7812-voltage regulator should be connected. (3)

5.2 What should be done with the unused input of a CMOS integrated circuit when it is connected to a circuit? (2)

[5]

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QUESTION 6: ELECTRONIC PHASE CONTROL

6.1 Make a neat, labelled sketch of a trigger circuit of a silicon-controlled rectifier. (2)

6.2 Show, with the aid of a sketch of a waveform, the operation of a phase-triggering circuit that is used in a single-phase controlled circuit. (4)
[6]

QUESTION 7: TEST EQUIPMENT

Complete the following sentences by writing only the missing term next to the question number (7.1–7.6) in the ANSWER BOOK.

The code register is converted to a/an (7.1) … reference voltage V_r which is compared with the (7.2) ... voltage V_u. The (7.3) … then enables the logic programmer to program the register with a certain (7.4) … Initially the most significant bit is set to a 1, while the other bits are 0. The (7.5) … analogue voltage V_r of this code is (7.6) … with Vu. If V_r < Vu this 1 is kept in the code, but if V_r > Vu the 1 is reset to a 0. **[6]**

QUESTION 8: OSCILLATORS

8.1 A piezoelectric crystal may have two resonant frequencies.

Make neat, labelled sketches of these TWO conditions and give a brief explanation. (5)

8.2 Briefly explain the function of the Rozner modification. (2)

8.3 Draw a neat, labelled circuit diagram which can be used to view level discrimination. (4)
[11]

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QUESTION 9: TRANSDUCERS

9.1 Calculate the resolutions of a wire-wound potentiometer with 350 turns. (2)

9.2 Choose a term from the following list to complete the sentences below. Write only the answer next to the question number (9.2.1–9.2.10) in the ANSWER BOOK.

greater; indorsed; phase; magnetic flux; size; secondary; displacement; indication; core; difference; oppose; proportional; external; induced

With the core in the centre, equal voltages will be (9.2.1) … in the secondary windings. Because they oppose each other, the output V_O will be 0V. As soon as an (9.2.2) … force moves the core to the left A, more (9.2.3) … develops in the left winding than the right winding (B). Therefore the (9.2.4) … voltage in winding A will be (9.2.5) … than that in winding B. Therefore the (9.2.6) … of the output voltage is equal to the (9.2.7) … between the two (9.2.8) … voltages of winding A. V_O is proportional to the amount of (9.2.9) … and the (9.2.10) … of V_O is an indication of the direction in which it moves. (10 × ½) (5)

9.3 Draw a neat, labelled circuit diagram of a photoelectric transducer that can be used for solar heat devices or satellite generators. (5)

[12]

TOTAL: 100

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INDUSTRIAL ELECTRONICS N5

FORMULA SHEET

$$I = \frac{V}{R} \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC \qquad T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt} \qquad V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL \qquad X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L \qquad Z = R - jX_C$$

$$Z = R + j(X_L - X_C) \qquad I_T = \frac{V_T}{Z_T}$$

$$V_R = I_T R \qquad V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C) \qquad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1 \qquad \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2} \qquad I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L} \qquad \frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L \qquad I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C} \qquad \frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C \qquad I_T = \frac{V}{R} + j\frac{V}{X_C}$$

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$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C) \qquad I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$$a + jb = \sqrt{a^2 + b^2} \,/\, \underline{\tan^{-1}\frac{b}{a}} = r\,/\underline{\theta}$$

$$r/\underline{\theta} = r(\cos\theta + j\sin\theta) \qquad Q = \tan\theta$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \qquad Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}}V_m = 0{,}707\,V_m \qquad V_{dc} = \frac{2}{\pi}V_m = 0{,}637\,V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P} \qquad V_{dc} = \frac{1}{\pi}V_m = 0{,}318\,V_m$$

$$PIV = V_m \qquad PIV = 2\,V_m$$

$$R_{r(rms)} = 0{,}385\,V_m \qquad V_{r(rms)} = 0{,}305\,V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}} \qquad V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2fC} \qquad V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L} \qquad V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L} \qquad r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L}.V_{dc} \qquad V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}}.V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R}.V_{r(rms)} \qquad r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$$

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$V'_{dc} = V_{dc} - I_{dc}R_1$

$V'_{dc} = \frac{R_L}{R_L + R_1} . V_{dc}$

$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$

$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$

$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$

$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$

$2V_m = V_{c2} = V_m + V_{c1}$

$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$

$S = \frac{\Delta V_o}{\Delta V_i}$

$V_R = V_i - V_z$

$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$

$I_z = \frac{P_z}{V_z}$

$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} . R_S$

$V_o = V_r - V_{be}$

$R_c = \frac{V_{cc} - V_{ce}}{I_c}$

$R_b = \frac{V_{cc} - V_{be}}{I_b}$

$\beta = \frac{I_c}{I_b}$

$C_e \geq \frac{10}{2\pi f R_e}$

$V_e = \frac{V_{cc}}{10}$

$R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$

$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$

$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$

$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$

$R_{b2} = \frac{1}{10} \beta R_e$

$V_b = V_e + V_{be}$

$V_{be} = h_{ie} i_b + h_{re} V_{ce}$

$i_c = i_{fe} i_b + h_{oe} V_{ce}$

$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$

$A_i = h_{fe}$

$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L} \right) \left(\frac{R_b T}{R_{bT} + Z_1} \right) \left(\frac{R_c}{R_c + R_L} \right)$

$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$

$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$

$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L}$

$Z_1 = h_{ie}$

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$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe}h_{re}}{h_{ie} + R_s}} \qquad Z_2 = \frac{1}{h_{oe}}$$

$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i \qquad A_p = \frac{{h_{fe}}^2 R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2 \qquad Z_1 = R_b // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1 \qquad I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe}\left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}}\right) \qquad A_i = \frac{I_0}{I_1}$$

For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.

$$Z_L = R_c // R_L \qquad I_1 = \frac{R_e I_i}{R_e + Z_1}$$

$$CMRR = \frac{A_{dm}}{A_{cm}} \qquad CMRR\ (dB) = 20log\frac{A_{dm}}{A_{cm}}$$

$$I_e = \frac{V_e}{R_e} \qquad I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C} \qquad g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left(\frac{R_f}{R_1}\right) . V_i \qquad V_0 = \left(\frac{R_f}{R_1} + 1\right) . V_i$$

$$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3\right) \quad V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3) \qquad V_0 = -(I_1 + I_2 + I_3)\, R_f$$

$$V_0(t) = -\frac{1}{RC}\int V_i(t) \qquad V_0(t_b) = -\frac{1}{RC}\int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

$$t = \frac{1}{f} \qquad A_v = -\frac{R_s}{R_1}$$

$$R_2 = \frac{R_1 R_s}{R_1 + R_s} \qquad f_c = \frac{1}{2\pi R_s C}$$

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$$V_0(t) = -RC\frac{dV_i(t)}{dt}$$

$$A = \frac{R_f}{X_c}$$

$$A = -\frac{R_f}{R_s}$$

$$V_0(t) = -R_f C\frac{d}{dt}.v_i \sin \omega t$$

$$t = R_f C$$

$$V_0 = A(V_r - V_i)$$

$$V_0 = \frac{R_f}{R_s}(V_2 - V_1)$$

$$V_0 = V_2 - V_1$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$$

$$L_T = L_1 + L_2 + 2M$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$$

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

$$f_0 = \frac{1{,}5}{RC}$$

$$f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$$

$$t_1 = 0{,}7\ R_2 C_1$$

$$t_2 = 0{,}7\ R_1 C_2$$

$$f_0 = \frac{1}{1{,}4RC}$$

$$V_i = I_{c2}R_e + V_{be(ON)}$$

$$t = 1{,}1\ RC$$

$$f_0 = \frac{1{,}443}{(R_A + 2R_B)C}$$

$$t_{low} = 0{,}693\,(R_B)C$$

$$t_{high} = 0{,}693\,(R_A + R_B)C$$

$$t_T = t_{low} + t_{high}$$

$$K = \frac{\Delta R / R}{\Delta l / l}$$

$$\sigma = \Delta l / l$$

$$R = \rho\frac{1}{\pi d^2 / 4}$$

$$\sigma = \frac{S}{E}$$

$$Resolution = \frac{1}{amount\ of\ turns}$$

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$$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$$

$R_t = Ae^{B/T}$ $\quad T = 273 + °C$

$V_A = \frac{R_2}{R_1 + R_2}.V_T$ $\quad V_B = \frac{R_t}{R_t + R_3}.V_T$

$V_{AB} = V_A - V_B$ $\quad A_v = \frac{V_0}{V_i}$

$V_{Hall} = kIH$

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